PROBLEM VF19

Viscous Pump with a Moving Rod

Original statement · VF19.pdf

Annulus | Moving rod | Flux and minimum velocity

OFFICIAL PROBLEM STATEMENT

Problem 4: To fill an open container of depth H located at a height L above the surface of a reservoir we use the viscous pumping system sketched in the figure, including a pipe or radius R_2 connecting the surface of the reservoir with the bottom of the container and an infinitely long, inner, coaxial rod of radius R_1 that moves vertically with velocity U, dragging the water into the inter-cylinder gap.

1. Give the condition for the flow in the gap to be dominated by viscosity.

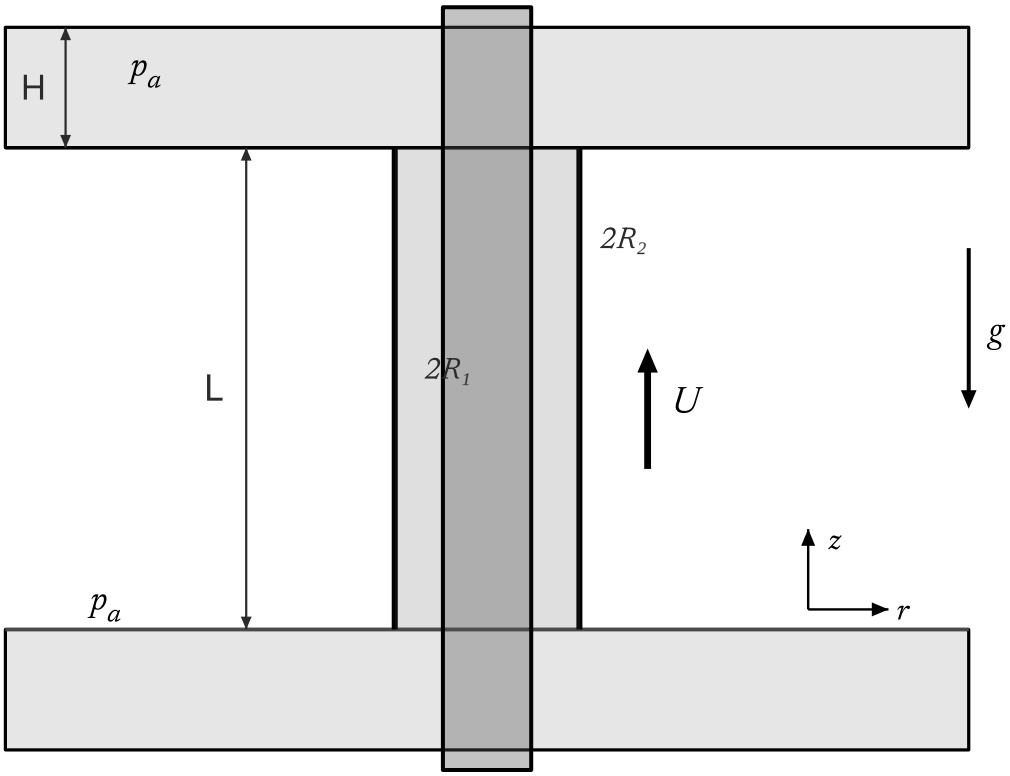
2. Obtain the velocity profile $v_z(r)$ across the gap.

3. Determine the loss of reduced pressure per unit length $P_l = -\partial P/\partial z$.

4. Compute the force acting on the moving shaft and the power needed to move it.

5. Determine the volume flow rate Q of the pumping system, including the minimum velocity U_{min} required for $Q > 0$.

The sheet prints "a pipe or radius R_2" ("or" for "of").



Statement figure (redrawn from VF19.pdf): a reservoir at p_a, a vertical pipe of radius R_2 and length L up to the bottom of an open container of depth H, and a coaxial rod of radius R_1 moving up with velocity U inside the pipe; z is vertical and r radial

Source: VF19.pdf (page 1, final exam of Jan 21, 2014; statement and handwritten key). Notes.pdf, Chapter 7: steady flow in a circular pipe, Eqs. 7.13-7.16; slender flows, Eqs. 7.40-7.46.

Worked solution

Hypotheses, Data and Unknowns

- The gap between the rod ($r = R_1$, moving with $+U\bar{e}_z$) and the pipe ($r = R_2$, at rest) has the length L and is slender, $R_2 - R_1 \ll L$; the flow is steady and axisymmetric, $v_z(r)$.
- z is the height above the reservoir surface. At the entrance of the pipe ($z = 0$) $p = p_a$; at its exit ($z = L$, the bottom of the container, under a depth H of water at p_a on top) $p = p_a + \rho g H$. The reduced pressure is $P = p + \rho g z$.

Symbol	Meaning	SI unit
R_1, R_2	radii of the rod and of the pipe	m

Symbol	Meaning	SI unit
L, H	length of the pipe, depth of the container	m
U	velocity of the rod (upward)	m/s
$\Delta = R_2^2 - R_1^2$, $\Lambda = \ln(R_2/R_1)$	abbreviations	m^2, 1

Criterion for Viscous Dominance

The slender-flow condition (Eq. 7.41), with the width $R_2 - R_1$ of the gap, the velocity U and the length L:

$$\frac{\rho U\,(R_2 - R_1)}{\mu}\,\frac{R_2 - R_1}{L} = \frac{\rho U\,(R_2 - R_1)^2}{\mu L} \ll 1$$

(the same with the velocity driven by the pressure, of order $P_l (R_2 - R_1)^2/\mu$, if it is larger than U).

Velocity Profile

Equation (7.7) in steady state with no slip on both walls:

$$0 = P_l + \frac{\mu}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right); \qquad r = R_1:\ v_z = U, \qquad r = R_2:\ v_z = 0$$

As in VF3, the solution is $v_z = \frac{P_l}{4\mu}\left(R_2^2 - r^2\right) + C\ln\frac{R_2}{r}$, which is zero at $r = R_2$; the rod condition gives $U = \frac{P_l}{4\mu}\Delta + C\Lambda$:

$$v_z = \frac{P_l}{4\mu}\left(R_2^2 - r^2\right) + \left[U - \frac{P_l}{4\mu}\left(R_2^2 - R_1^2\right)\right]\frac{\ln(R_2/r)}{\ln(R_2/R_1)}$$

Pressure Loss per Unit Length

The gap has constant section, so P_l is uniform. The reduced pressure at the entrance is p_a and at the exit $p_a + \rho g H + \rho g L$:

$$P_l = -\frac{\partial P}{\partial z} = \frac{p_a - [p_a + \rho g (H + L)]}{L} = -\frac{\rho g (H + L)}{L}$$

negative: the pressure (reduced) increases along the flow, opposing it; the rod has to push the water against the weight of the whole column from the reservoir to the top of the container.

Force on the Rod and Power

The force of the liquid on the rod (normal from the rod into the liquid, $\bar{n} = +\bar{e}_r$) per unit area is $\tau_w = \mu \, \partial v_z / \partial r |_{r=R_1}$. With $\frac{dv_z}{dr} = -\frac{P_l}{2\mu} r - \frac{C}{r}$:

$$\tau_w = -\frac{P_l R_1}{2} - \frac{\mu C}{R_1} = -\frac{\mu U}{R_1 \Lambda} - \frac{P_l R_1}{2} \left[1 - \frac{1}{2} \, \frac{R_2^2 / R_1^2 - 1}{\ln (R_2 / R_1)} \right]$$

(using $C = \left(U - P_l \Delta / (4\mu) \right) / \Lambda$). With $P_l = -\rho g (H + L) / L$ both terms are negative: the liquid resists the motion of the rod, because $\frac{\Delta}{2 R_1^2 \Lambda} = \frac{x^2 - 1}{2 \ln x} > 1$ for $x = R_2 / R_1 > 1$. The force on the rod over the length L is $F_z = 2\pi R_1 L \, \tau_w \, \bar{e}_z$, downward, and the power that must be supplied to move it is minus the power of that force (the key writes $W = U \, 2\pi R_1 L \, \tau_w$, the power of the force of the liquid, negative):

$$W = -2\pi R_1 L U \tau_w = 2\pi R_1 L U \left[\frac{\mu U}{R_1 \ln (R_2 / R_1)} + \frac{\rho g (H + L) R_1}{2L} \left(\frac{R_2^2 - R_1^2}{2 R_1^2 \ln (R_2 / R_1)} - 1 \right) \right]$$

The key's lines end at the expression of τ_w without writing the final value.

Volume Flux and Minimum Velocity

$Q = \int_{R_1}^{R_2} 2\pi r\, v_z\, dr$. Barrow's rule for the two terms:

$$\int_{R_1}^{R_2} r\left(R_2^2 - r^2\right) dr = \left[\frac{R_2^2 r^2}{2} - \frac{r^4}{4}\right]_{R_1}^{R_2} = \frac{\left(R_2^2 - R_1^2\right)^2}{4}, \qquad \int_{R_1}^{R_2} r \ln\frac{R_2}{r}\, dr = \left[\frac{r^2}{2}\ln\frac{R_2}{r} + \frac{r^2}{4}\right]_{R_1}^{R_2} = \frac{R_2^2 - R_1^2}{4} - \frac{R_1^2}{2}\ln\frac{R_2}{R_1}$$

$$Q = \frac{\pi P_l \Delta^2}{8\mu} + \frac{\pi}{2}\left(U - \frac{P_l \Delta}{4\mu}\right)\frac{\Delta - 2R_1^2\Lambda}{\Lambda}$$

Discrepancy with the official solution

The key gives $Q = -2\pi\left\{\frac{P_l}{16\mu}\left(R_2^4 - R_1^4\right) + \frac{U - P_l\Delta/(4\mu)}{4\ln(R_2/R_1)}\left[R_2^2\left(2\ln R_2 - 1\right) - R_1^2\left(2\ln R_1 - 1\right)\right]\right\}$, which is not the integral of its own velocity profile. The first term should have $\left(R_2^2 - R_1^2\right)^2$ (the integral of $r(R_2^2 - r^2)$, as above), not $R_2^4 - R_1^4$; and the second keeps only the antiderivative of $r\ln r$ and drops the constant term that comes from $\ln R_2$ (a logarithm of a dimensional length appears, which is dimensionally meaningless); the overall sign is also the opposite. As a consequence the key's minimum velocity, solved from its own $Q = 0$, is not the one of the integral. The results above are used.

Checks of Q: for $R_1 \to 0, \Delta \to R_2^2, R_1^2\Lambda \to 0, \Lambda \to \infty$ and the second term vanishes: $Q \to \frac{\pi P_l R_2^4}{8\mu}$, the Poiseuille flux (7.15). For a thin gap $g = R_2 - R_1 \ll R_1$, $(\Delta - 2R_1^2\Lambda)/\Lambda \simeq 2R_1 g \cdot \frac{g}{R_1} \cdot R_1$ (to first order $2R_1^2\epsilon$ with $\epsilon = g/R_1$), so the second term is $\pi R_1^2 \epsilon U = \pi R_1 g U$, the Couette flux $\frac{Ug}{2} \cdot 2\pi R_1$.

Minimum velocity. Setting $Q = 0$: $\left(U - \frac{P_l\Delta}{4\mu}\right) = -\frac{P_l\Delta^2}{4\mu}\frac{\Lambda}{\Delta - 2R_1^2\Lambda}$, so

$$U_{min} = \frac{P_l\Delta}{4\mu}\left[1 - \frac{\Delta\,\Lambda}{\Delta - 2R_1^2\Lambda}\right] = \frac{\rho g(H+L)\left(R_2^2 - R_1^2\right)}{4\mu L}\left[\frac{\left(R_2^2 - R_1^2\right)\ln(R_2/R_1)}{R_2^2 - R_1^2 - 2R_1^2\ln(R_2/R_1)} - 1\right]$$

positive, since $\frac{\Delta\Lambda}{\Delta - 2R_1^2\Lambda} > 1$. For $U > U_{min}$ the rod pumps water to the container ($Q > 0$).

Answers and Checks

- Viscous dominance: $\rho U(R_2 - R_1)^2/(\mu L) \ll 1$.
- Velocity: $v_z = \frac{P_l}{4\mu}(R_2^2 - r^2) + \left[U - \frac{P_l}{4\mu}(R_2^2 - R_1^2)\right] \frac{\ln(R_2/r)}{\ln(R_2/R_1)}$; $P_l = -\rho g(H + L)/L$.
- Force on the rod: $F_z = 2\pi R_1 L \tau_w$ with τ_w above; power supplied: $W = -2\pi R_1 L U \tau_w > 0$.
- Flux: $Q = \frac{\pi P_l \Delta^2}{8\mu} + \frac{\pi}{2}\left(U - \frac{P_l \Delta}{4\mu}\right) \frac{\Delta - 2R_1^2\Lambda}{\Lambda}$; U_{min} as above.
- Dimensions: $\rho g(H + L)\Delta/(\mu L)$ is $ML^{-3} \cdot LT^{-2} \cdot L^2/(ML^{-1}T^{-1}) = LT^{-1}$, a velocity; $\Delta^2 P_l/\mu$ is $L^4 \cdot ML^{-2}T^{-2}/(ML^{-1}T^{-1}) = L^3T^{-1} \cdot L^{-1}$, a flux per unit length times L (as in (7.15)).
- Check against the official key: VF19.pdf gives the same criterion, profile, P_l and shear; its Q and U_{min} differ, as noted, and it does not write the final value of the power.